

SHAHD JBAREEN

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EDUCATION

Sami Shamoon College of Engineering

Be'er Sheva, Israel *B.S. Software Engineering*

Expected Graduation: June 2026

- **Relevant Coursework:** Data Structures, Algorithms I–II, Databases, Operating Systems, Computer Architecture I–II, Computer Communications, Computer Communication Networks, Data Security, Object-Oriented Programming, Advanced OOP, Software Engineering, Requirements Engineering, Software Testing and Quality, Compilation, Parallel and Distributed Computing.

EXPERIENCE

Final Project Developer / Research Software Developer | Volcani Institute, Rishon LeZion

Oct. 2025 – Present

- Developing AgroMind, an interactive AI-based decision support system and Android mobile application for smart irrigation and nitrogen fertilization.
- Designed farmer-facing mobile features, including daily recommendations, confidence scores, Gemini-based explanations, and AI Plan vs. User Plan comparisons.
- Integrated AquaCrop simulation and Genetic Algorithm optimization outputs into a clear mobile workflow using Rolling Window / MPC-like planning.
- Contributed to what-if analysis, result validation, quality assurance testing, technical documentation, and preparation of an international research article.

Web and Application Development Trainee | Local Office, Umm al-Fahm

2024 – Present

- Gained practical experience in developing websites and applications according to client and business requirements.
- Assisted in implementing user interfaces, application logic, database-related functionality, and improvements for existing digital systems.

SKILLS

Programming Languages: Java, Python, JavaScript, Kotlin, C/C++ , C, SQL

Mobile Development: Android Studio, Jetpack Compose, Retrofit, OkHttp, Firebase

Back-End Technologies: ASP.NET, Django, FastAPI/Flask, REST APIs, WebSocket, Microservices

AI and Optimization: Gemini API, OpenAI GPT, Genetic Algorithms, Rolling Window / MPC-like Optimization, TensorFlow, Scikit-learn

Databases and Tools: MySQL, SSMS, PostgreSQL, SQLite, Git/GitHub, Jenkins, Jira, SSL Certificates

System Architecture: API Design, Data Processing, System Scaling, Performance Optimization, Security Enhancement

PROJECTS

CampusFlow – Smart Academic Request Management System | Python, Django, SQLite, HTML, CSS, JavaScript, 2025

Jira, GitHub, Jenkins, OpenAI API

- Developed a role-based academic request management platform for students, lecturers, secretaries, and administrators, supporting request submission, status tracking, approval workflows, and dashboards.
- Implemented authentication, profile management, admin user management, reports, notifications, and AI chatbot support for student guidance.
- Worked with Jira and GitHub for Agile sprint planning, task tracking, version control, and team collaboration.
- Used Jenkins for CI workflows and contributed to unit and integration testing across user stories and system roles.

Compiler Construction Project | C, Lex, Yacc, Jenkins, AST, Symbol Table, Three-Address Code

2025

- Developed a compiler for a custom procedural programming language, including lexical analysis, parsing, AST construction, semantic analysis, and intermediate code generation.
- Implemented semantic checks using a symbol table, including scope validation, function and variable declarations, type checking, return validation, and parameter matching.
- Generated Three-Address Code for valid programs, including control-flow handling for conditional statements, loops, function calls, and short-circuit logical expressions.
- Used Jenkins to support automated build and testing workflows during the project development process.

eBook Library Service | C, HTML, CSS, JavaScript, ASP.NET, SSMS, PayPal, SSL

2025

- Developed an eBook library management system with user roles, borrowing limits, catalog management, and payment processing.

- Integrated PayPal API and SSL certificates to support secure transactions and managed database functionality using SSMS.

Blockchain Project for Real-Time Data Processing | *Blockchain, Smart Contracts, Python* 2024

- Built a backend system for decentralized and secure data management using blockchain concepts, smart contracts, and Python processing.

Children's Game App | *C, Python, Game Development* 2024

- Developed an educational game application using C and Python, focusing on interactive features and backend functionality.

LANGUAGES

Arabic – Native, Hebrew – Fluent, English – Intermediate